Shubhadeep Mandal

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Education

International Institute of Information Technology, Hyderabad — July 2025 – Present
Telangana, India
Master of Technology in Computer Science and Engineering — *CGPA: 9.12*

Indian Institute of Information Technology, Ranchi — Aug 2020 – Jun 2024
Ranchi, India
Bachelor of Technology (Hons.) in Computer Science and Engineering — *CGPA: 9.05*

Projects

RookDB: Vectorized Relational Storage Engine | *Rust, Database Internals, B+Tree, CBO, Volcano*

- Co-architected a multi-crate SQL-99 RDBMS in Rust verified by **700+ value-level tests**, engineering slotted-page storage, CLOCK buffer pools, B+Tree indexing, and Volcano query execution.
- Engineered vectorized Volcano query operators, a Cost-Based Optimizer with HyperLogLog statistics, and Grace Hash Join, executing **2M-match joins in 5.85s** and streaming scans at **16.9M rows/s**.
- Designed an executor-tier cache and LRU query plan cache, eliminating 8 per-row I/O bottlenecks and driving a **100–400x DML throughput turnaround** (38,986 inserts/sec vs. 94 baseline) at 10M-row scales.

Peer-to-Peer Distributed File Sharing System | *C++, POSIX Sockets, Multithreading, OpenSSL, Linux*

- Engineered a BitTorrent-style hybrid P2P system in C++, achieving an **8.5x transfer speedup on 1GB files** (193s to 22.8s) across 5 concurrent peers by saturating network and disk I/O.
- Implemented out-of-order parallel chunk writing directly to designated file offsets and dynamic bitfield generation for real-time partial seeding during active downloads.
- Architected Primary-Backup tracker failover with 2s bounded failover, full state synchronization, and verified resilience via **48 unit tests** under AddressSanitizer/UBSan.

IndicLLM: Transformer Pretraining & Reasoning SFT | *PyTorch, FlashAttention-2, RoPE*

- Pretrained 25M-param decoder-only Transformers from scratch on 1B+ tokens across Hindi and Assamese using bare PyTorch primitives, driving a +36.5% perplexity reduction via FlashAttention-2, RoPE, and SwiGLU.
- Trained dedicated 16k SentencePiece BPE tokenizers on AI4Bharat Sangraha, eliminating token fertility inflation and achieving 0.505 Bits-Per-Byte sequence compression across both high and low-resource languages.
- Fine-tuned 8 model variants across 20k synthetic multi-hop reasoning tasks, unlocking a +45.8% relative Token F1 gain and 61.3% decomposed CoT step score over direct SFT via intermediate deductive scratchpads.

FOOL: Custom POSIX-Compliant Interactive Shell | *C++, POSIX APIs, Job Control, Termios, Linux*

- Built an interactive POSIX shell in C++ supporting multi-stage pipelines, I/O redirections, and wildcard expansion, validated via an automated **98-check test suite**.
- Eliminated signal race conditions and lost exit statuses using the self-pipe trick for async-safe `SIGCHLD` handling and centralizing process reaping on the main thread.
- Accelerated startup by **52x** (5.2s to 0.1s on WSL) via directory entry caching; built zero-dependency raw terminal editor using POSIX `termios` with tab autocompletion and persistent history.

Technical Skills

Languages: C/C++, Rust, Python, TypeScript, JavaScript, SQL, Bash.

Systems & OS: POSIX APIs, Sockets, Linux/Unix, Multithreading, Signals, Termios, OpenSSL, Storage Engines, ASan/UBSan.

AI & Machine Learning: Custom Transformers (Pretraining & SFT), FlashAttention-2, RoPE, SwiGLU, RMSNorm, CoT, PyTorch, SentencePiece.

Tools & Cloud: Docker, Git, CI/CD, Next.js, React, Convex DB, PostgreSQL, WebSockets.

Achievements

- **B.Tech Rank Holder:** Ranked 2nd in IIIT Ranchi by GPA in B.Tech CSE.
- **PGEE 2025:** Secured All-India Rank 100 in IIIT-Hyderabad Post Graduate Entrance Examination (PGEE).
- **Megathon 2025:** Third rank team in Megathon 2025 in the mobile computing track.

Interests

- Web Novel and Manga Enthusiast.
- Chess and Badminton.